

Deep Dive: 5. Control Flow

Concept Summary

Control flow allows your program to make decisions and execute specific blocks of code based on conditions.

Coding Structure

```
if condition:
    # code block
elif condition:
    # code block
else:
    # code block
```

Example Code

```
score = 85
if score >= 90:
    print('Grade: A')
elif score >= 80:
    print('Grade: B')
else:
    print('Grade: C')
```

Code Explanation

- Line 2: Evaluates if score is 90 or above. It is not, so it skips to the elif.
- Line 4: Evaluates if score is 80 or above. Since $85 \geq 80$ is True, it executes the indented block underneath it.
- Line 5: Prints 'Grade: B'. The else block is completely ignored.

Practice Problems

1. Write a script that checks a temperature variable and prints 'Wear a coat' if it's below 50, otherwise 'T-shirt time'.
2. Create a nested if-statement that checks if a user is an adult (≥ 18) and, if so, whether they are a senior (≥ 65).